

# FIREWALL INTEGRATION & MONITORING REPORT

## DEPLOYMENT ARCHITECTURE:

| Component                        | Host                | IP Address                                           | Operating System |
|----------------------------------|---------------------|------------------------------------------------------|------------------|
| Wazuh-Server                     | VMware              | 192.168.18.119                                       | Ubuntu 24.04 LTS |
| Windows Agent                    | Host Windows        | 192.168.18.106                                       | Windows 11 Pro   |
| Linux-EndpointFW-Test            | VMware              | 192.168.50.45                                        | Ubuntu 24.04 LTS |
| OPNsense Firewall                | VMware              | WAN:<br>192.168.18.127/24<br>LAN:<br>192.168.50.1/24 | OPNsense 26.7.1  |
| Validation Ubuntu VM (Non-Admin) | VMware (Old Laptop) | 192.168.50.128                                       | Ubuntu 24.04 LTS |

## NETWORK SETUP:

- Wazuh Server and Windows Agent were configured on the **bridged network (192.168.18.0/24)** to enable communication with the local network and Wazuh Dashboard.
- OPNsense Firewall was configured with two network interfaces:
  - WAN Interface:** Bridged network (192.168.18.128/24) for Internet connectivity.
  - LAN Interface:** Host-Only network (192.168.50.0/24) to provide an isolated internal network for endpoint testing.
- The **Linux-Endpoint-FW-Test** and **Validation Ubuntu VM (Non-Admin)** were connected to the OPNsense LAN network (192.168.50.0/24) and received IP addresses through the OPNsense DHCP server.
- OPNsense was configured as the **default gateway** and **DNS server** for all devices on the LAN network, ensuring that all traffic passed through the firewall before reaching the Internet.
- Internet connectivity from the Linux endpoint was successfully verified through the OPNsense firewall.

- The **Wazuh Agent** was installed and configured on the OPNsense firewall to enable centralized security monitoring from the Wazuh Dashboard.
- OPNsense was successfully enrolled as an active Wazuh agent (**Agent ID: 005**).
- Communication between the OPNsense firewall and the Wazuh Server was established through the Wazuh Agent using the default Wazuh agent-manager communication channel.
- Wazuh Dashboard was accessible at <https://192.168.18.119>, where the OPNsense firewall and other monitored endpoints were successfully displayed.

## OPNsense FIREWALL INSTALLATION:

**Step 1: Download OPNsense ISO** OPNsense Community Edition ISO was downloaded from the official website:

<https://opnsense.org/download/>



The screenshot shows the OPNsense download page with three main sections: System architecture, Select the image type, and Mirror Location. The System architecture dropdown is set to 'amd64'. The Select the image type dropdown is set to 'dvd'. The Mirror Location dropdown is set to 'OPNsense'. Below the Select the image type dropdown, there is a description: 'dvd: ISO installer image with live system capabilities running in VGA mode. On amd64, UEFI boot is supported as well.' Below this description is a 'Download OPNsense®' button. To the right of the Mirror Location dropdown, there is a note: 'OPNsense can be downloaded from a large range of mirrors located in different countries, you may want to select the fastest options for your location.'

## Step 2: Create OPNsense Virtual Machine

A new virtual machine was created in VMware Workstation Pro with the following specifications:

- Operating System: FreeBSD 64-bit
- Memory: Minimum 2 GB RAM
- Processor: 2 vCPUs
- Disk: 20 GB (thin provisioned)
- Network Adapter 1: Bridged (WAN)
- Network Adapter 2: Host-Only / Custom (LAN)

| Device            | Summary                          |
|-------------------|----------------------------------|
| Memory            | 2.5 GB                           |
| Processors        | 2                                |
| Hard Disk (SCSI)  | 25 GB                            |
| CD/DVD (IDE)      | Using file C:\Users\w t\Downl... |
| Network Adapter   | Bridged (Automatic)              |
| Network Adapter 2 | Host-only                        |
| USB Controller    | Present                          |
| Sound Card        | Auto detect                      |
| Display           | Auto detect                      |

## Step 3: Install OPNsense

The OPNsense ISO was booted and the installation process was initiated.

- Boot from ISO
- Select "Install" from the boot menu
- Select keyboard layout: US
- Select ZFS or UFS filesystem (UFS selected for simplicity)
- Partitioning: Guided (Entire Disk)
- Set root password: [Password saved securely]
- Reboot after installation

#### Step 4: Assign Network Interfaces

During first boot, OPNsense prompted for interface assignment:

| Interface | Assignment      | Network                        |
|-----------|-----------------|--------------------------------|
| WAN       | em0 (or vtnet0) | Bridged Network (DHCP)         |
| LAN       | em1 (or vtnet1) | Host-Only<br>(192.168.50.1/24) |

The interfaces were verified using the console menu option 1 (Assign Interfaces).

```
*** OPNsense-FW2.internal: OPNsense 26.7.1 (amd64) ***

LAN (em1)      -> v4: 192.168.50.1/24
WAN (em0)      -> v4/DHCP4: 192.168.18.128/24

HTTPS: SHA256 B6 68 65 F4 C3 DB 4E 7C 10 0B EC D6 A1 B5 4D B7
              6F 49 5C CB 44 02 F8 5A 4D FF 82 31 6F 7A AA 1A

0) Logout                      7) Ping host
1) Assign interfaces           8) Shell
2) Set interface IP address    9) pfTop
3) Reset the root password     10) Firewall log
4) Reset to factory defaults   11) Reload all services
5) Power off system            12) Update from console
6) Reboot system               13) Restore a backup

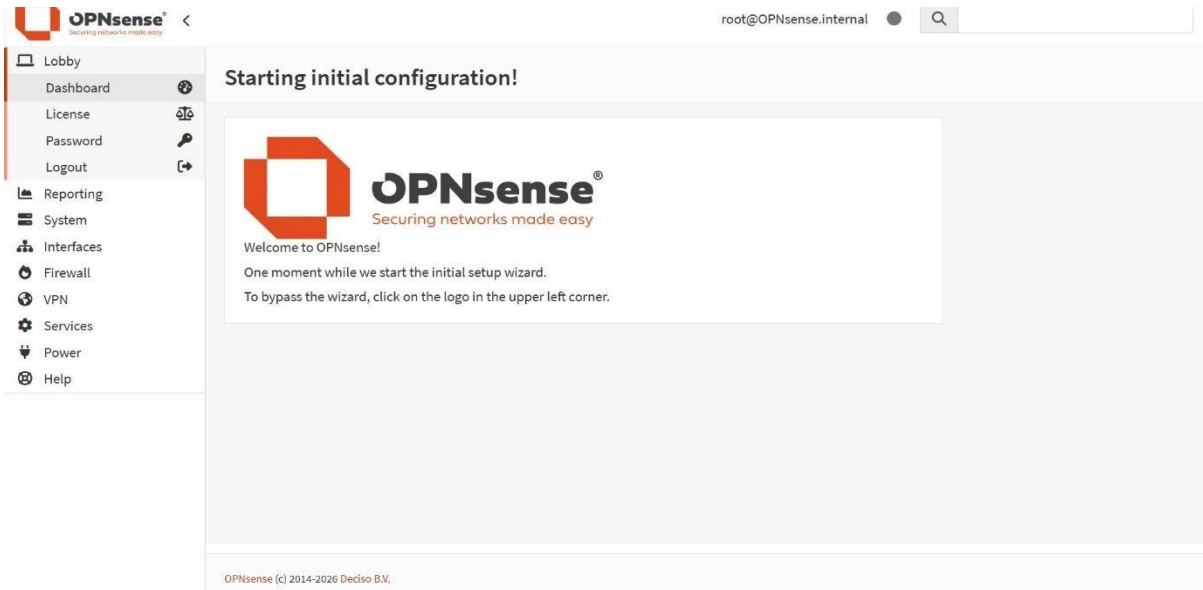
Enter an option: █
```

#### Step 5: Access OPNsense Web GUI

After interface assignment, the Web GUI was accessed via HTTPS:

- URL: <https://192.168.50.1>
- Default Username: root
- Default Password: opnsense **NOTE:**

The default password was changed immediately after first login for security.



## INTERFACE CONFIGURATION:

### WAN Interface:

| Setting            | Value                                 |
|--------------------|---------------------------------------|
| Interface          | em0 (vtnet0)                          |
| IPv4 Configuration | DHCP                                  |
| IPv6 Configuration | Disabled                              |
| Status             | Up                                    |
| IP Address         | 192.168.18.128/24 (Assigned via DHCP) |

Welcome   General Information   **Network [WAN]**   Network [LAN]   Deployment type

---

1 Disable WAN ☐

1 Type

3 MAC (spoofed)

1 MTU

1 MSS

1 DHCP hostname

✓ Default policies

1 Block RFC1918 Private Networks ☐

1 Block bogon networks ☐

Next

### LAN Interface:

| Setting   | Value        |
|-----------|--------------|
| Interface | em1 (vtnet1) |

|                           |                 |
|---------------------------|-----------------|
| <b>IPv4 Configuration</b> | Static IPv4     |
| <b>IPv4 Address</b>       | 192.168.50.1/24 |
| <b>IPv6 Configuration</b> | Disabled        |
| <b>Status</b>             | Up              |

Welcome
General Information
Network [WAN]
Network [LAN]
Deployment type

Disable LAN ☐

IP Address 192.168.100.1/24

Configure DHCP server ☒

Next

*\*192.168.100.1/24 was later changed to 192.168.50.1/2\**

## DHCP CONFIGURATION:

DHCP Server was configured on the LAN interface to automatically assign IP addresses to test clients:

| Setting            | Value                                          |
|--------------------|------------------------------------------------|
| <b>Interface</b>   | LAN                                            |
| <b>DHCP Server</b> | Enabled                                        |
| <b>Range</b>       | 192.168.50.100 - 192.168.50.200                |
| <b>DNS Servers</b> | 192.168.50.1 (OPNsense DNS Forwarder/Resolver) |
| <b>Gateway</b>     | 192.168.50.1                                   |
| <b>Lease Time</b>  | 24 hours                                       |

Domain local

Language English

Timezone Asia/Karachi

DNS Servers 8.8.8.8 1.1.1.1

Clear All Copy Paste Text

Override DNS ☐

DNS [Unbound]

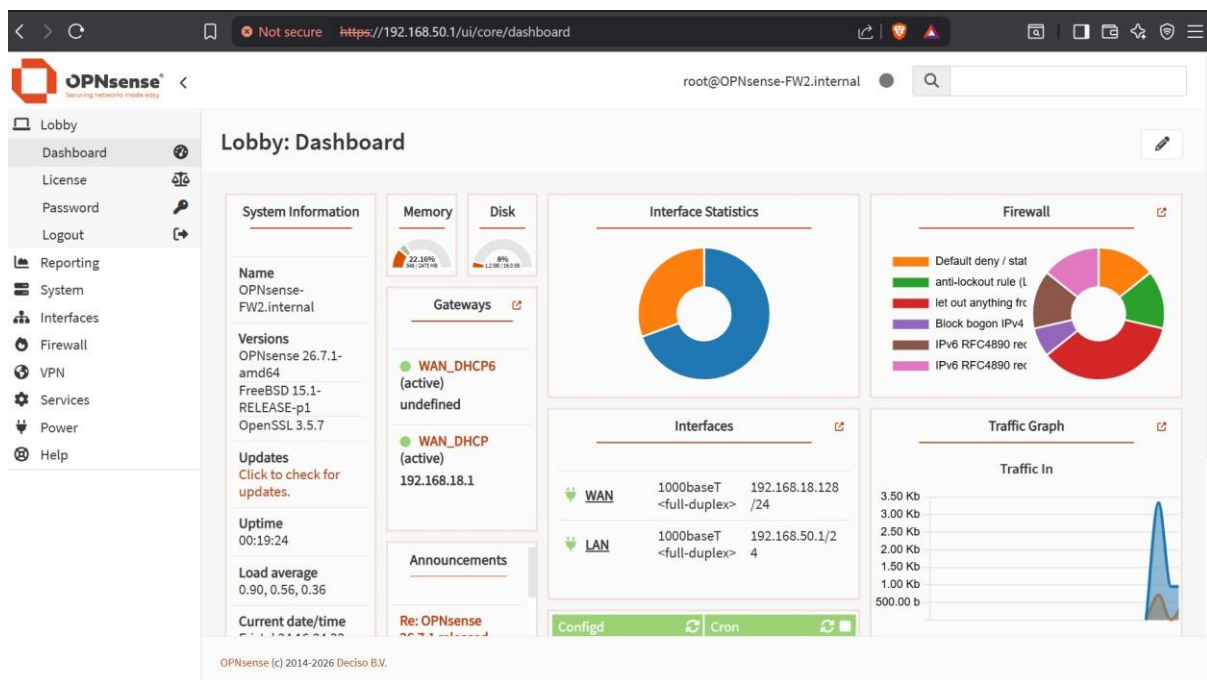
Enable Resolver ☒

Enable DNSSEC Support ☒

Harden DNSSEC data ☒

*\*8.8.8.8 & 1.1.1.1 was later changed to 192.168.50.1 because Linux endpoint was configured to use OPNsense as their DNS server\**

## OPNsense Dashboard after configurations:



## WAZUH AGENT INTEGRATION

### Step 1: Install the Wazuh Agent on OPNsense

System → Firmware → Plugins

The Wazuh Agent plugin was installed on the OPNsense firewall to enable centralized security monitoring from the Wazuh Server. The plugin was installed through the OPNsense package manager and configured to communicate with the Wazuh Manager.

The screenshot shows the 'System: Firmware' page with the 'Plugins' tab selected. It displays a table of installed plugins, including the Wazuh Agent.

| Status                     | Settings | Changelog | Updates | Plugins    | Packages                                                                                        |
|----------------------------|----------|-----------|---------|------------|-------------------------------------------------------------------------------------------------|
| wa                         | Version  | Size      | Tier    | Repository | Comment                                                                                         |
| os-wazuh-agent (installed) | 1.3_1    | 42.0KiB   | 3       | OPNsense   | Agent for the open source security platform Wazuh                                               |
| os-dec-hw                  | 1.1_3    | 6.87KiB   | 2       | OPNsense   | Deciso hardware specific information                                                            |
| os-sunnyvalley             | 1.5_2    | 2.43KiB   | 2       | OPNsense   | Vendor Repository for Zenarmor - Enterprise SASE & SSE platform (NGFW, SWG, CASB, ZTNA, SD-WAN) |
| os-vmware                  | 1.5_1    | 645B      | 2       | OPNsense   | VMware tools                                                                                    |

### Step 2: Configure Wazuh Agent

Services → Wazuh Agent → Settings

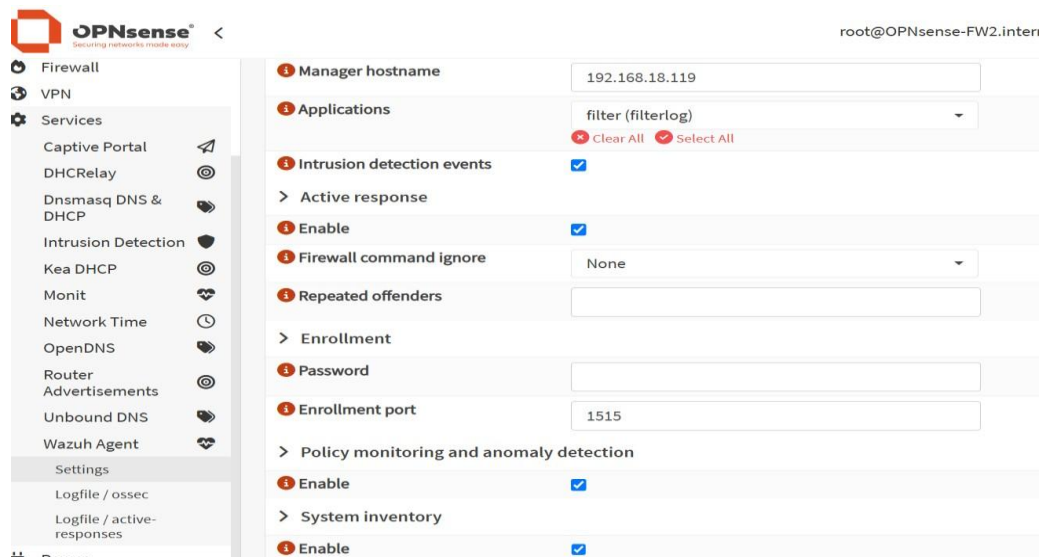
The Wazuh Agent was configured with the Wazuh Manager's IP address and authentication details.

### Configuration Details:

| Setting          | Value          |
|------------------|----------------|
| Wazuh Manager IP | 192.168.18.119 |

|              |          |
|--------------|----------|
| Manager Port | 1515/TCP |
| Registration | Enabled  |

After the configuration was completed, the Wazuh Agent service was started on the OPNsense firewall.



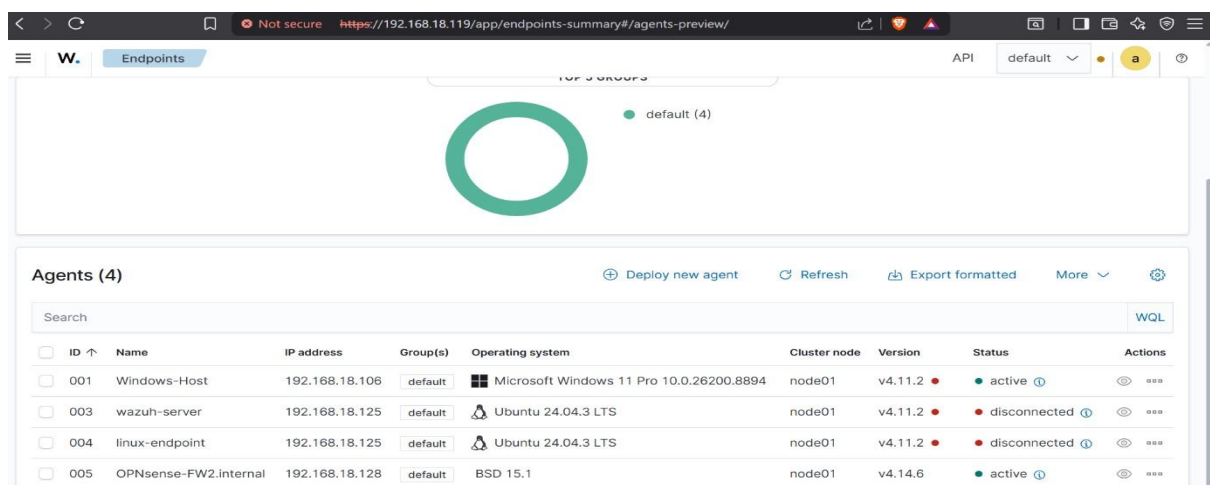
### Step 3: Register the OPNsense Firewall with the Wazuh Server

Wazuh Dashboard → Server Management → Endpoints (Agents)

The OPNsense firewall was successfully enrolled with the Wazuh Server. Once registration was completed, the firewall appeared as an active endpoint in the Wazuh Dashboard.

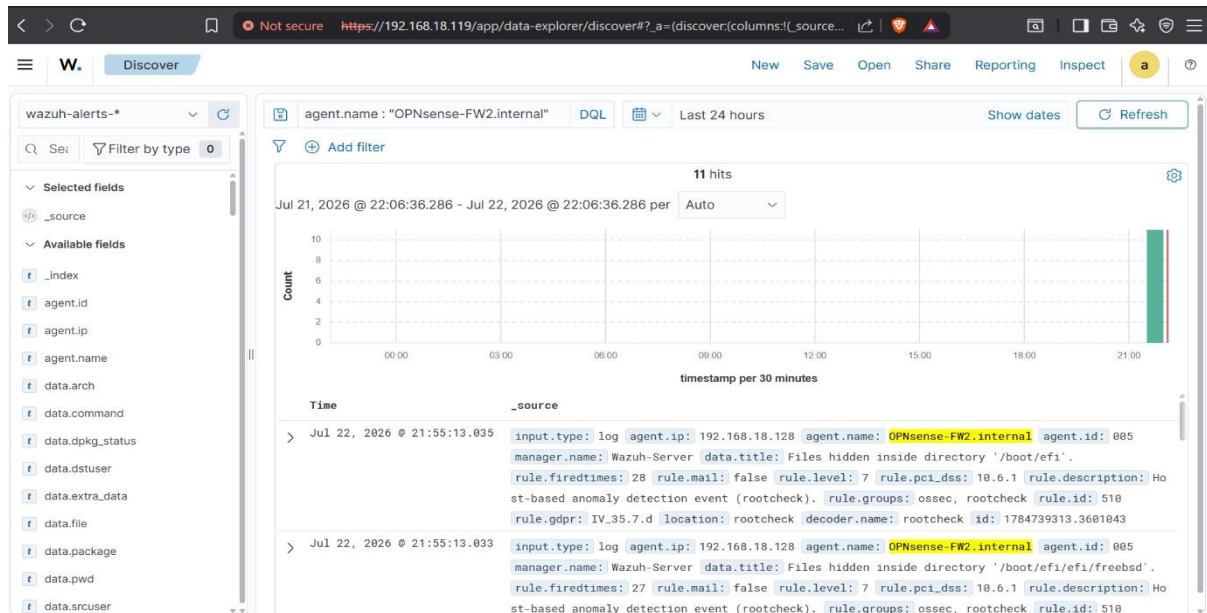
#### Agent Details

| Parameter | Value                 |
|-----------|-----------------------|
| Agent ID  | 005                   |
| Hostname  | OPNsense-FW2.internal |
| Status    | Active                |



## Step 4: Verify Successful Integration

The Wazuh Dashboard was used to verify successful communication between the OPNsense firewall and the Wazuh Server. The firewall was displayed as an active Wazuh agent, confirming successful enrollment and centralized monitoring. This integration enabled continuous monitoring of the OPNsense firewall from the Wazuh platform throughout the implementation and validation of the firewall security policies.



## Custom Wazuh Rule Creation:

A custom Wazuh detection rule was created to generate a dedicated alert whenever the OPNsense Wazuh Agent produced a Rootcheck event. The rule was implemented in the local\_rules.xml file on the Wazuh Manager.

Command: `sudo nano /var/ossec/etc/rules/local_rules.xml`

```
<!-- OPNsense Rootcheck Custom Detection -->
<rule id="110005" level="10">
  <if_sid>510</if_sid>
  <description>Custom: OPNsense rootcheck security event detected</description>
  <group>opnsense,rootcheck,custom</group>
</rule>
```

## Rule Validation:

The custom rule configuration was verified using the Wazuh rule validation utility, and the Wazuh Manager service was restarted to load the newly created rule successfully.



```

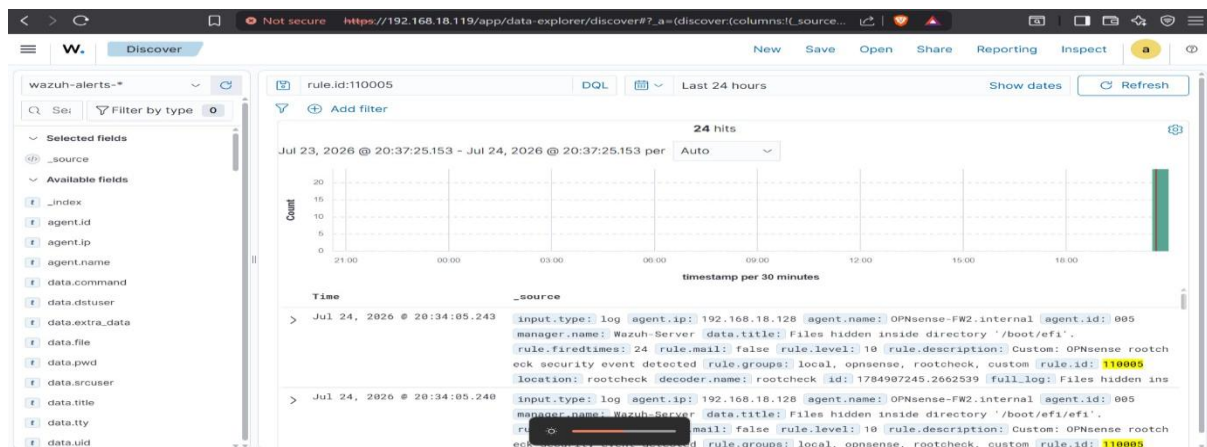
fara@Wazuh-Server:~/Desktop$ sudo systemctl status wazuh-manager --no-pager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-24 20:54:05 PKT; 4min 34s ago
     Invocation: 1b1fec3c36194f3cb40dd7f958aa4e62
   Process: 1635 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 233 (limit: 4570)
   Memory: 277M (peak: 1G, swap: 242M, swap peak: 242.1M)
      CPU: 2min 16.870s
   CGroup: /system.slice/wazuh-manager.service
           └─3224 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─3225 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─3226 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─3229 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─3232 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─3276 /var/ossec/bin/wazuh-authd
                       └─3292 /var/ossec/bin/wazuh-db
                         └─3329 /var/ossec/bin/wazuh-execd
                           └─3348 /var/ossec/bin/wazuh-analysisd
                             └─3362 /var/ossec/bin/wazuh-syscheckd
                               └─3382 /var/ossec/bin/wazuh-remoted
                                 └─3383 /var/ossec/bin/wazuh-remoted
                                   └─3478 /var/ossec/bin/wazuh-logcollector
                                     └─3493 /var/ossec/bin/wazuh-monitord
                                       └─3517 /var/ossec/bin/wazuh-modulesd

Jul 24 20:54:00 Wazuh-Server env[1635]: Started wazuh-logcollector...
Jul 24 20:54:00 Wazuh-Server env[1635]: wazuh-monitord: Process 7654 not used by Wazuh, removing...
Jul 24 20:54:01 Wazuh-Server env[1635]: Started wazuh-monitord...
Jul 24 20:54:01 Wazuh-Server env[1635]: wazuh-modulesd: Process 7663 not used by Wazuh, removing...
Jul 24 20:54:01 Wazuh-Server env[3514]: 2026/07/24 20:54:01 wazuh-modulesd:router: INFO: Loaded router module.
Jul 24 20:54:01 Wazuh-Server env[3514]: 2026/07/24 20:54:01 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Jul 24 20:54:01 Wazuh-Server env[3514]: 2026/07/24 20:54:01 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester module.
Jul 24 20:54:03 Wazuh-Server env[1635]: Started wazuh-modulesd...
Jul 24 20:54:05 Wazuh-Server env[1635]: Completed.
Jul 24 20:54:05 Wazuh-Server systemd[1]: Started wazuh-manager.service - Wazuh manager.

```

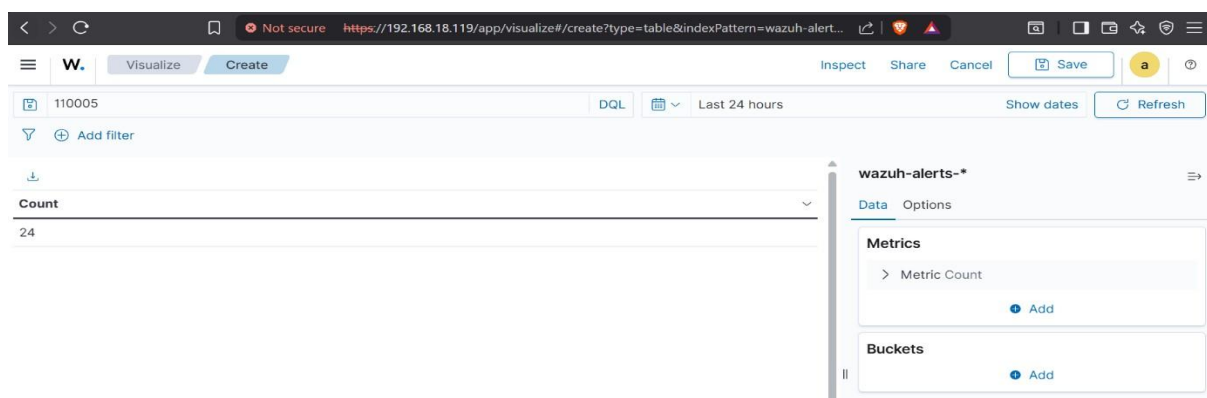
## Alert Verification:

After the rule was loaded, a new Rootcheck event generated by the OPNsense agent successfully triggered the custom Rule ID 110005. The alert appeared in the Wazuh Discover page with the custom description and alert level.



## Dashboard Monitoring:

A custom dashboard visualization was created using the Wazuh Alerts index to monitor alerts generated by Rule ID 110005. This dashboard provides a centralized view of custom Rootcheck detections associated with the OPNsense firewall.



## SECURITY POLICIES IMPLEMENTED:

### A. GeoIP Blocking (China, Russia)

#### Step 1: Configure GeoIP Database

##### Firewall → Aliases → GeoIP Settings

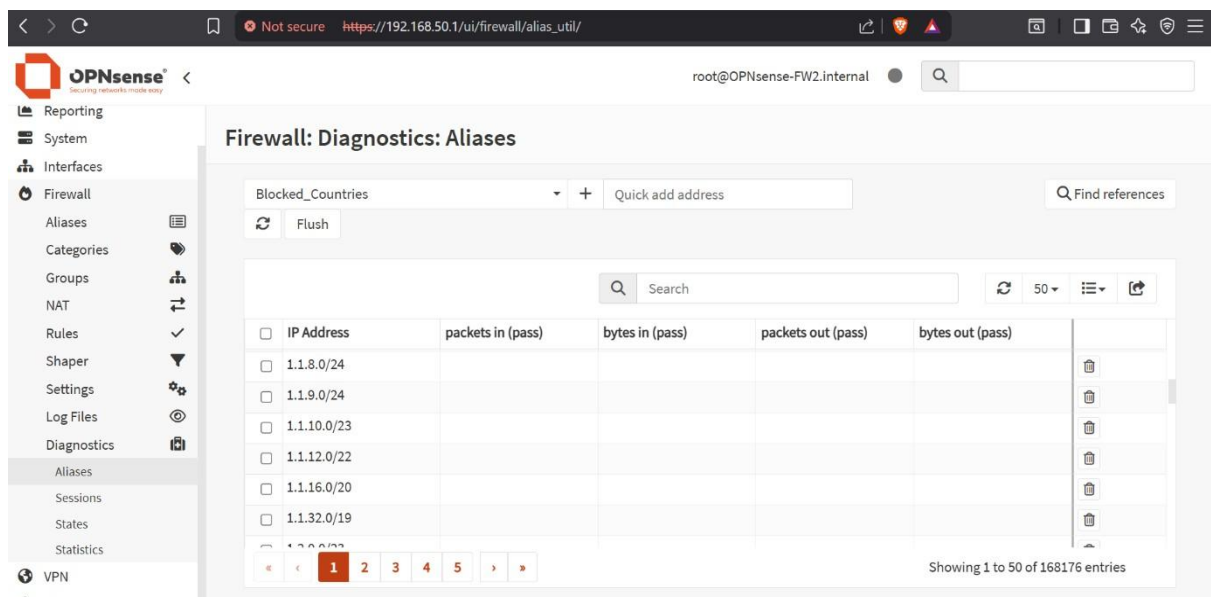
The built-in GeoIP feature in OPNsense was configured to download the IPinfo Lite Country database, which provides IP ranges mapped to their respective countries.

#### Configuration:

- Provider: **IPinfo Lite**
- Database Type: **Country CSV**
- Download Database



We can see blocked Ips in the following screenshot:



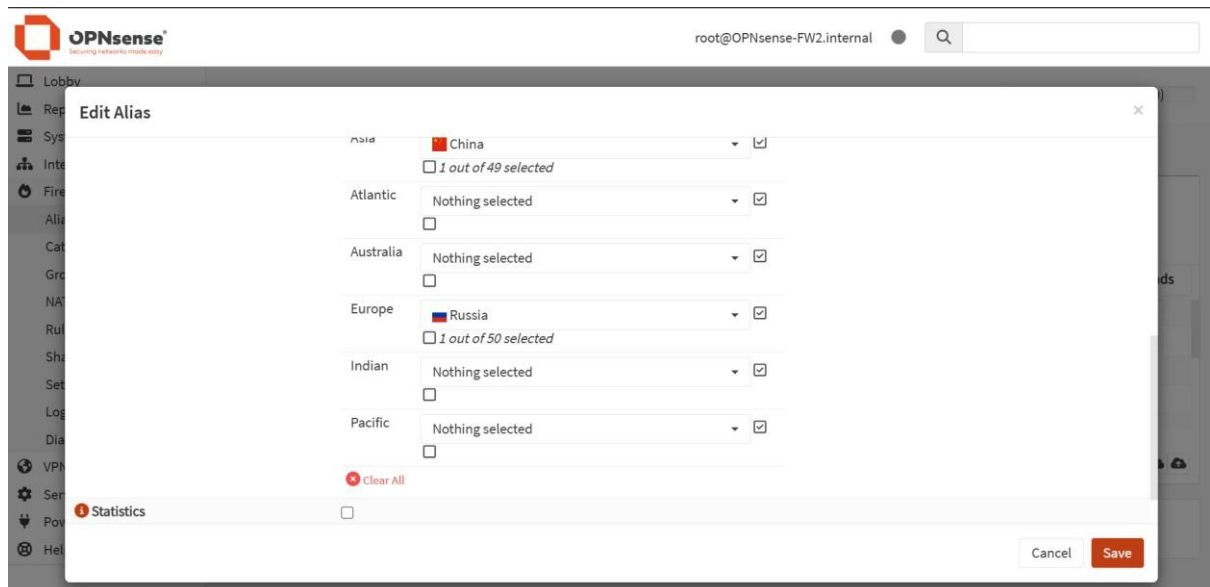
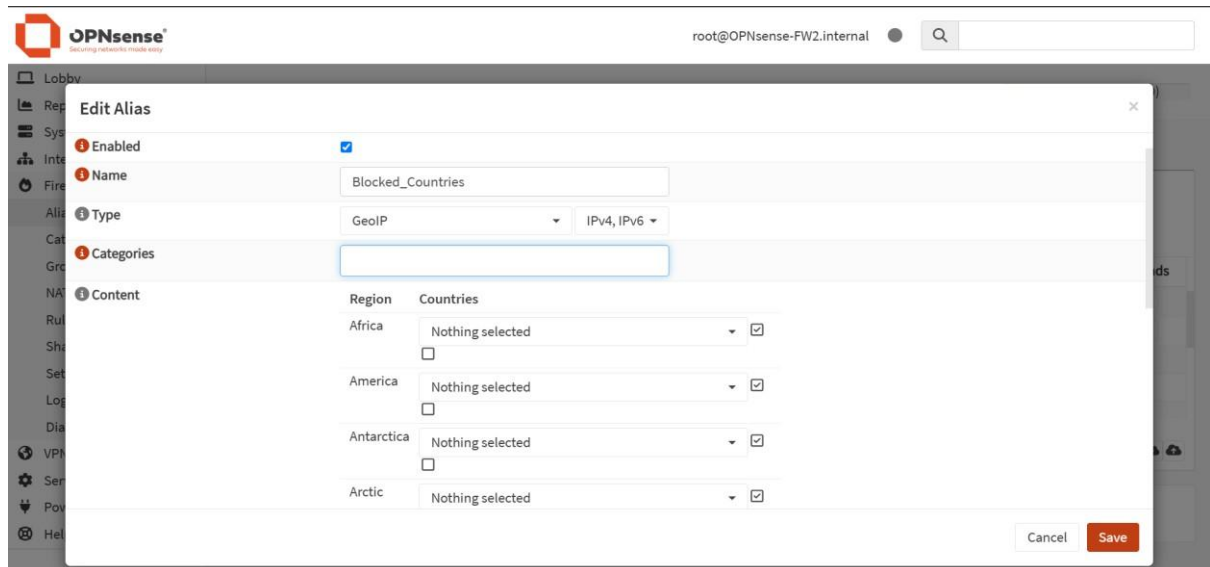
#### Step 2: Create GeoIP Alias

##### Firewall → Aliases → Add

A GeoIP alias containing the IP ranges for China and Russia was created to simplify firewall rule management.

### Configuration:

- Name: **Blocked\_Countries**
- Type: **GeoIP**
- Countries Selected: ☐ China (CN) ☐ Russia (RU)
- Description: **Countries to be blocked**



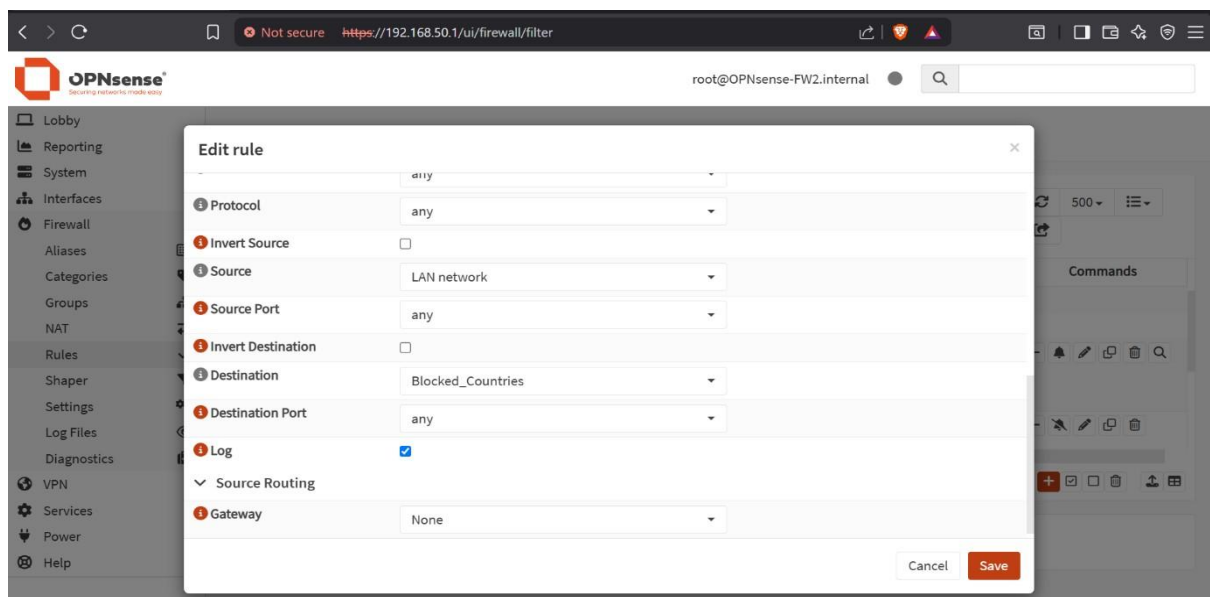
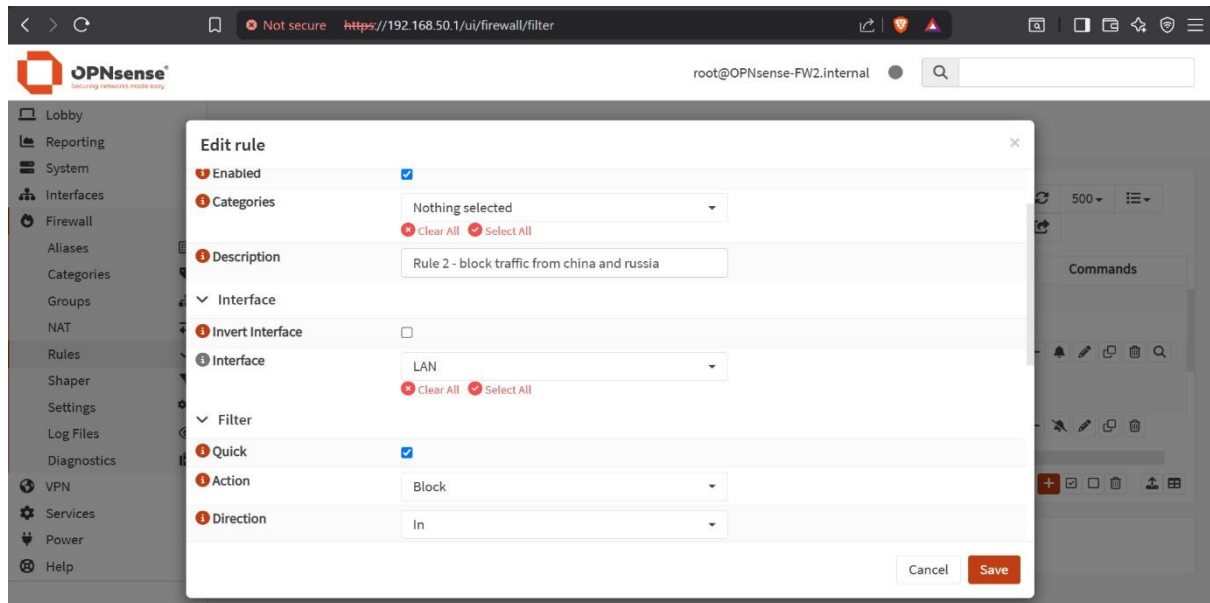
### Step 3: Create GeoIP Firewall Rule

Firewall → Rules → LAN → Add

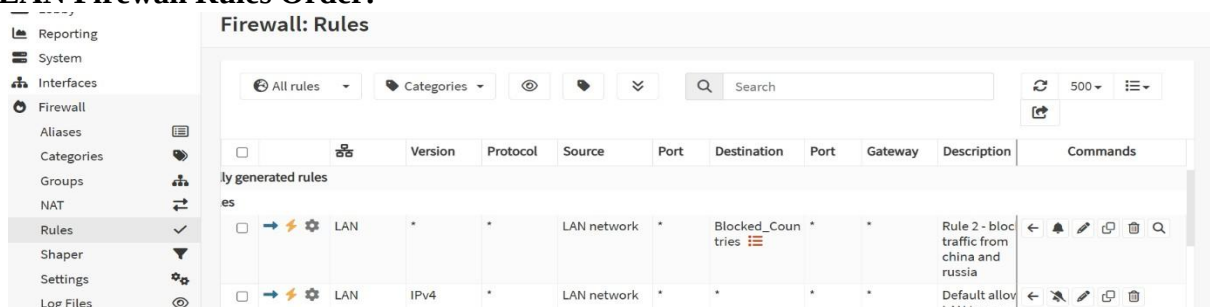
A firewall rule was created on the LAN interface to block outbound connections from internal devices to IP addresses belonging to the selected countries.

### Configuration:

- Action: **Block**
- Protocol: **Any**
- Source: **LAN Net**
- Destination: **Blocked\_Countries**
- Log: **Enabled**
- Description: **Block Traffic to China and Russia**



## LAN Firewall Rules Order:



## Step 4: Validate GeoIP Blocking

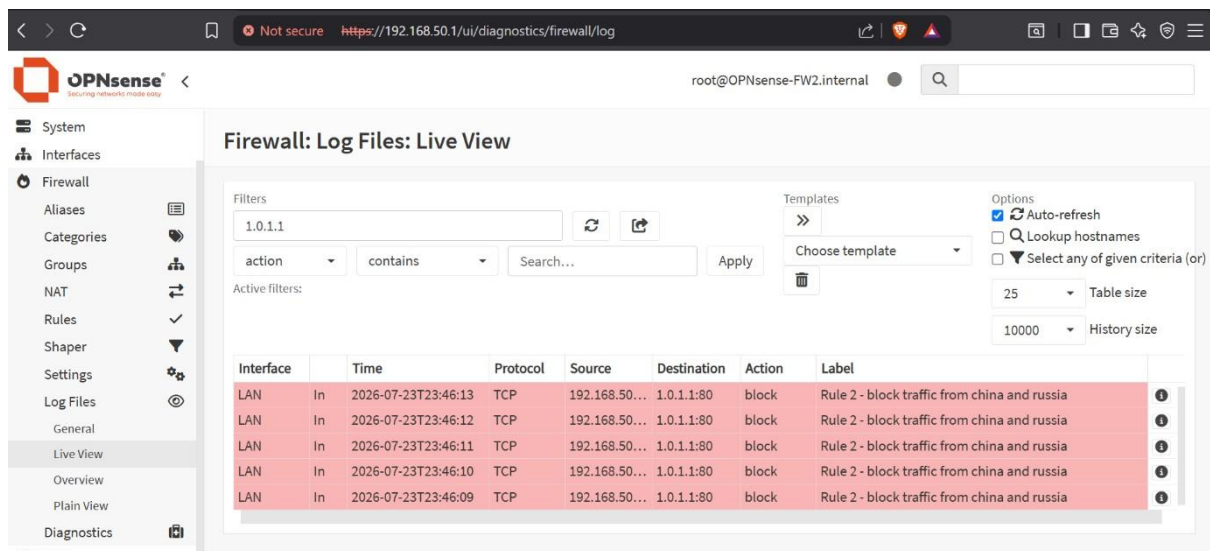
The GeoIP policy was validated by generating traffic from the Linux endpoint to an IP address contained within the GeoIP alias.

### Validation Performed:

- Source Host: **Linux-Endpoint-FW-Test**
- Tried connecting to 1.0.1.1
- Result: **Traffic was successfully blocked by the firewall.**
- Verification: The blocked connection was recorded in the **Firewall Live View**.

```
fara@Linux-Endpoint-FW-Test:~/Desktop$ curl --connect-timeout 5 http://1.0.1.1
curl: (28) Connection timed out after 5002 milliseconds
```

Firewall → Log Files → Live View



OPNsense Firewall: Log Files: Live View

Filters: 1.0.1.1

Templates: Choose template

Options: ☒ Auto-refresh, ☐ Lookup hostnames, ☐ Select any of given criteria (or)

Table size: 25, History size: 10000

| Interface | Time                   | Protocol | Source        | Destination | Action | Label                                        |
|-----------|------------------------|----------|---------------|-------------|--------|----------------------------------------------|
| LAN       | In 2026-07-23T23:46:13 | TCP      | 192.168.50... | 1.0.1.1:80  | block  | Rule 2 - block traffic from china and russia |
| LAN       | In 2026-07-23T23:46:12 | TCP      | 192.168.50... | 1.0.1.1:80  | block  | Rule 2 - block traffic from china and russia |
| LAN       | In 2026-07-23T23:46:11 | TCP      | 192.168.50... | 1.0.1.1:80  | block  | Rule 2 - block traffic from china and russia |
| LAN       | In 2026-07-23T23:46:10 | TCP      | 192.168.50... | 1.0.1.1:80  | block  | Rule 2 - block traffic from china and russia |
| LAN       | In 2026-07-23T23:46:09 | TCP      | 192.168.50... | 1.0.1.1:80  | block  | Rule 2 - block traffic from china and russia |

## B. Web Content Filtering

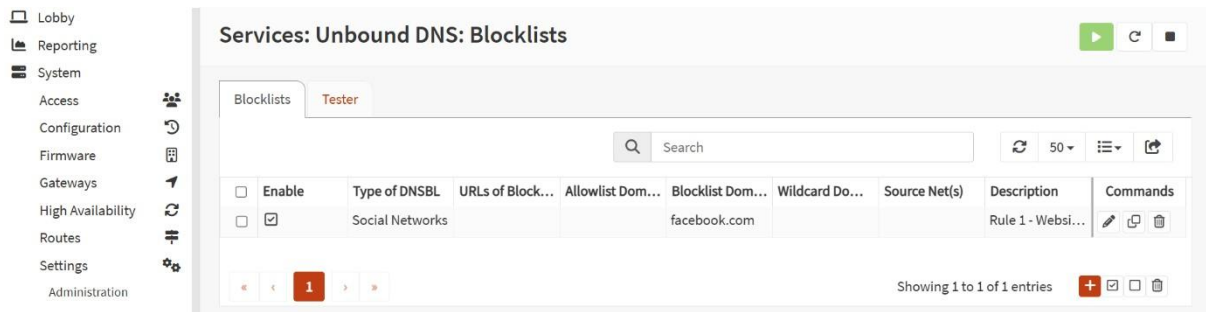
### Step 1: Configure Unbound DNS Block Lists

Services → Unbound DNS → Blocklists

The Unbound DNS Block Lists feature was configured to enable DNS-based web content filtering.

### Configuration:

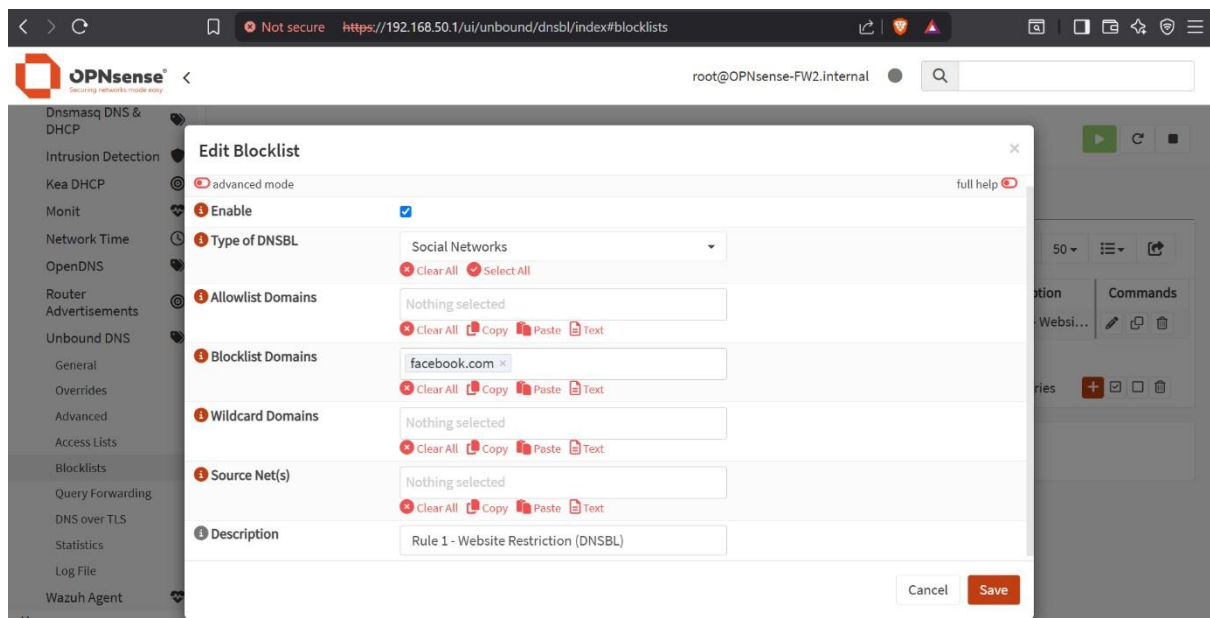
- Enable **Blocklists**
- Save the configuration



## Step 2: Add Website to Block List

Services → Unbound DNS → Blocklists

The target website was added to the DNS Block List to prevent users on the LAN network from resolving its domain name. Add facebook.com to blocked domain. Apply and save the configuration.



## Step 3: Apply Configuration

After adding the blocked domain, the Unbound DNS service was reloaded to apply the updated DNS Block List configuration.

Services → Unbound DNS → Blocklists → Apply

## Step 4: Validate Website Blocking

Website blocking was validated from the Linux endpoint.



```
fara@Linux-Endpoint-FW-Test:~/Desktop$ sudo resolvectl flush-caches
[sudo: authenticate] Password:
fara@Linux-Endpoint-FW-Test:~/Desktop$ dig facebook.com

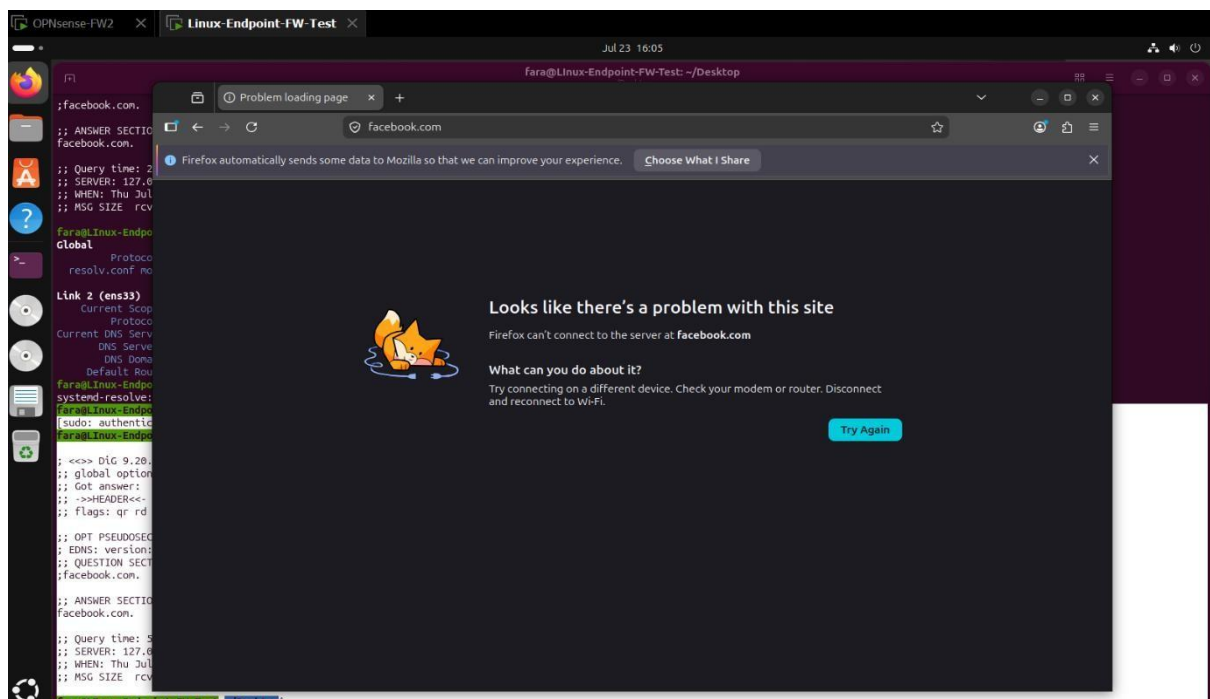
; <<>> DiG 9.20.18-ubuntu2.1-Ubuntu <<>> facebook.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48493
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;facebook.com.
                IN      A

;; ANSWER SECTION:
facebook.com.    3600    IN      A      0.0.0.0

;; Query time: 5 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jul 23 16:03:44 PKT 2026
;; MSG SIZE rcvd: 57
```

Accessing **facebook.com** from Firefox was unsuccessful, confirming that DNSBL was functioning correctly.



## C. Administrator Privilege and Access Control

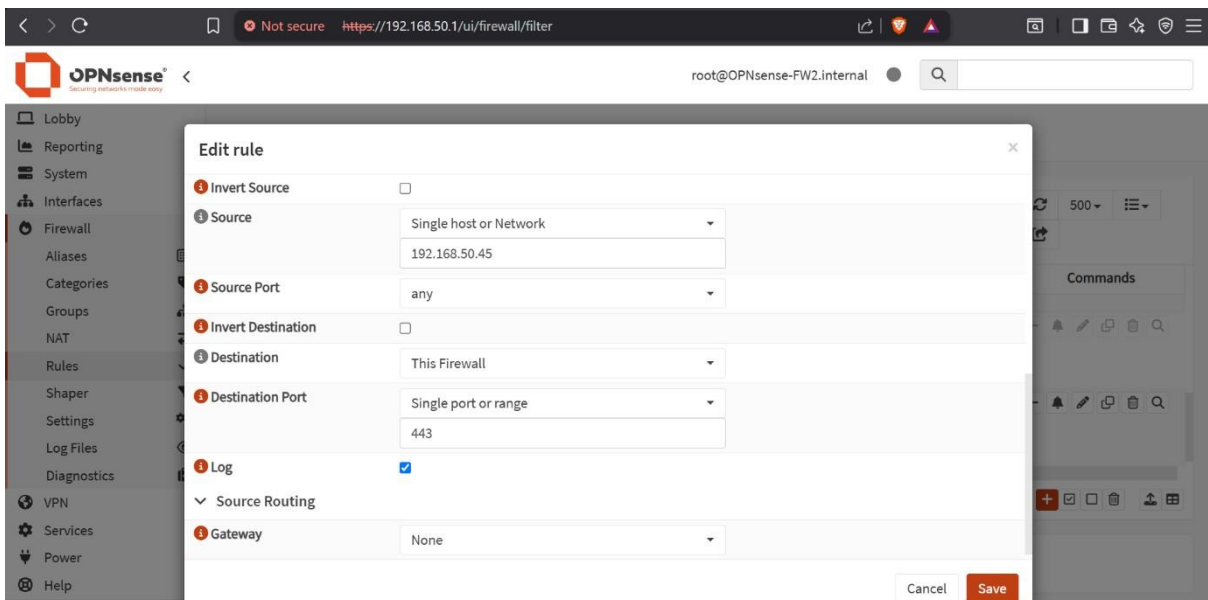
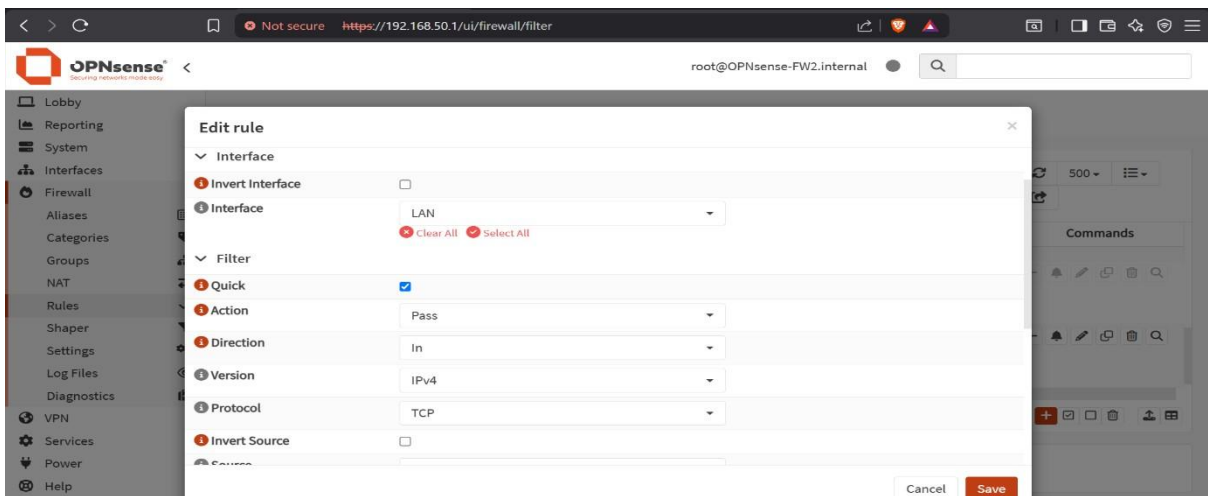
### Step 1: Create Administrator Access Rule

A firewall rule was created to allow only the authorized administrator workstation to access the OPNsense web management interface.

Firewall → Rules → LAN → Add

- Action: **Pass**
- Protocol: **TCP**
- Source: **192.168.50.45** (Linux-Endpoint-FW-Test)
- Destination: **This Firewall**

- Destination Port: **HTTPS (443)**
- Log: **Enabled**
- Description: **Allow Admin PC**



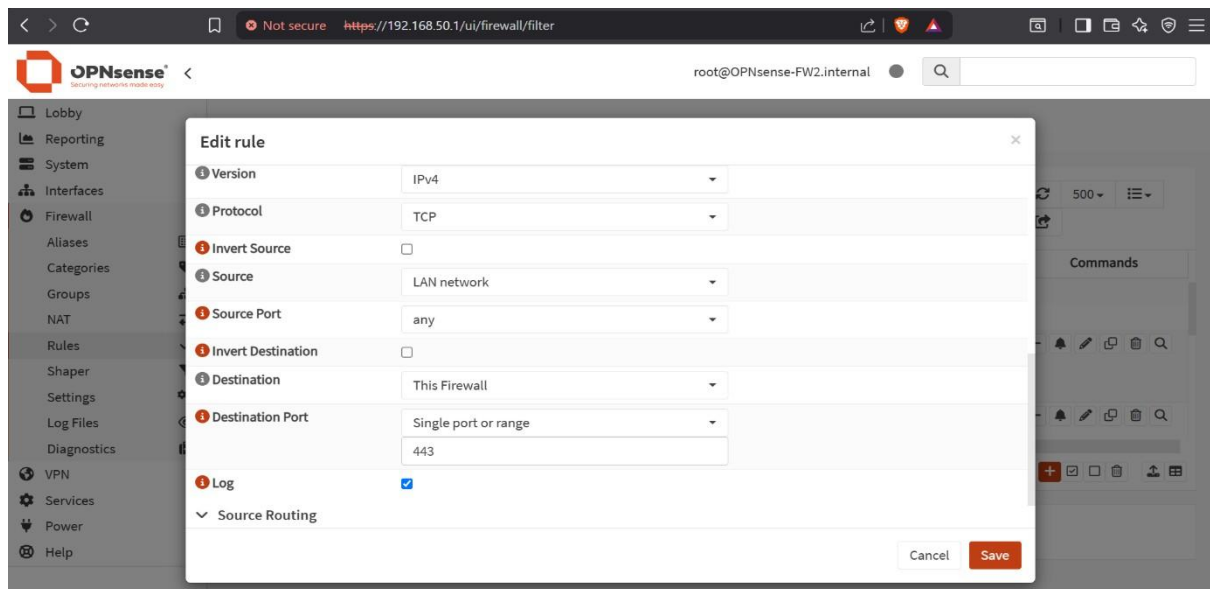
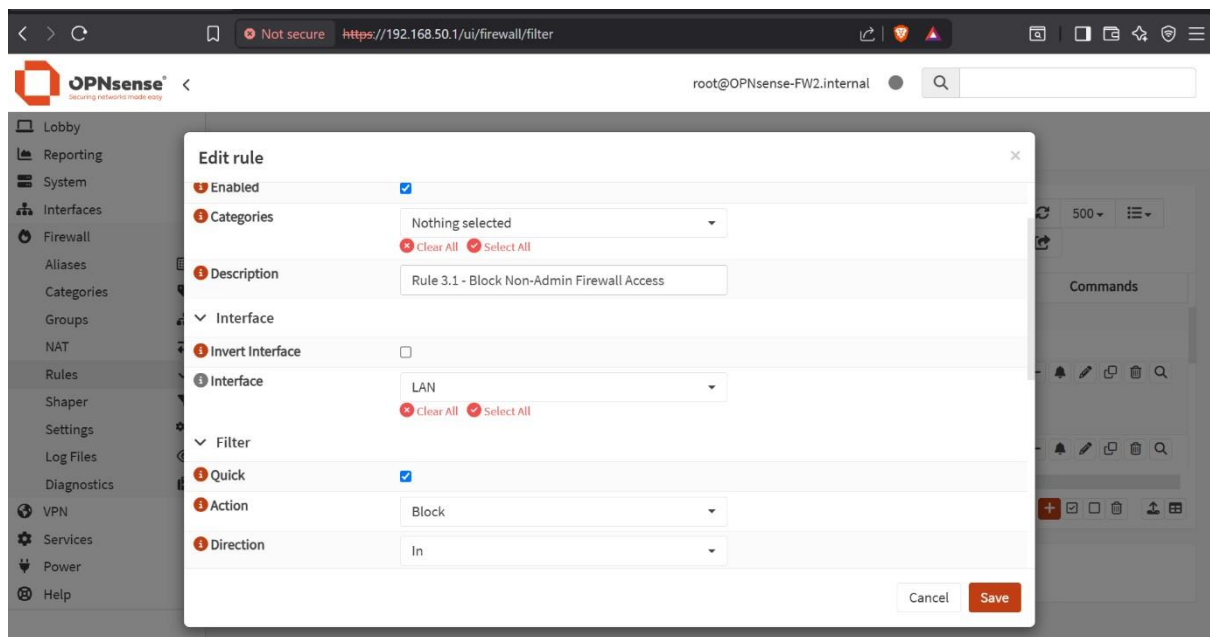
## Step 2: Create Non-Administrator Access Rule

A second firewall rule was created to deny management access to all other devices on the LAN.

Firewall → Rules → LAN → Add

- Action: **Block**
- Protocol: **TCP**
- Source: **LAN Net**
- Destination: **This Firewall**
- Destination Port: **HTTPS (443)**
- Log: **Enabled**
- Description: **Block Non-Admin Firewall Access**





### Step 3: Firewall Rule Order

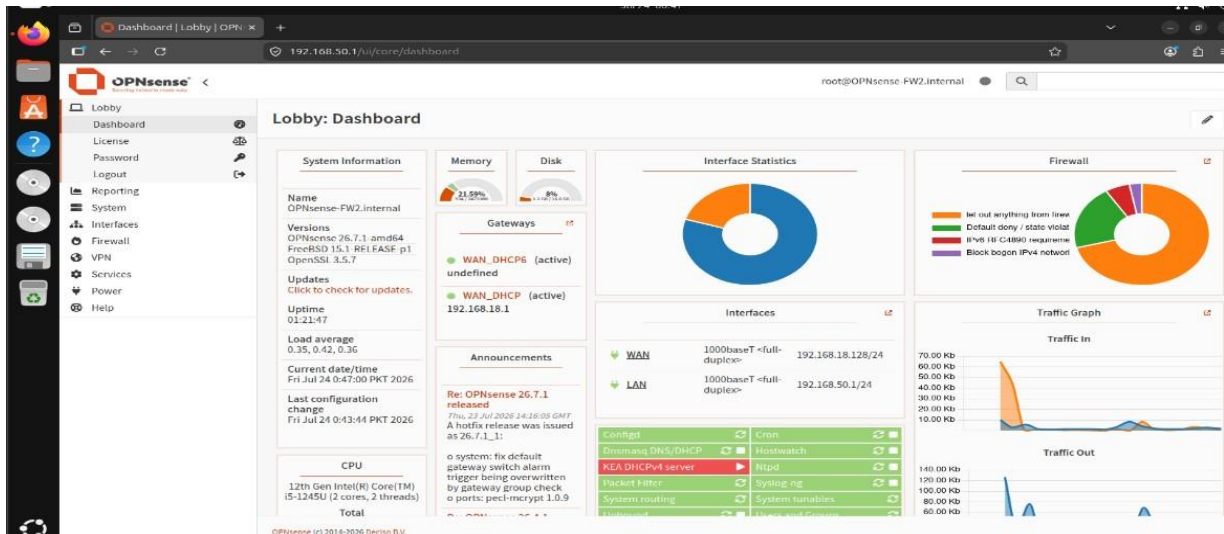
The administrator access rule was placed above the blocking rule to ensure that authorized access was permitted before the general block rule was evaluated.

1. Allow Admin PC to manage firewall
2. Block Non-Admin Firewall Access
3. Block Countries
4. Default Allow LAN IPv4
5. Default Allow LAN IPv6

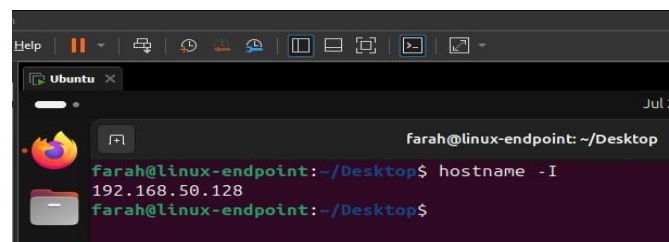
### Step 4: Validate Administrator Access Control

The administrator access control policy was validated using two separate systems to confirm that only the authorized administrator workstation could access the OPNsense web interface.

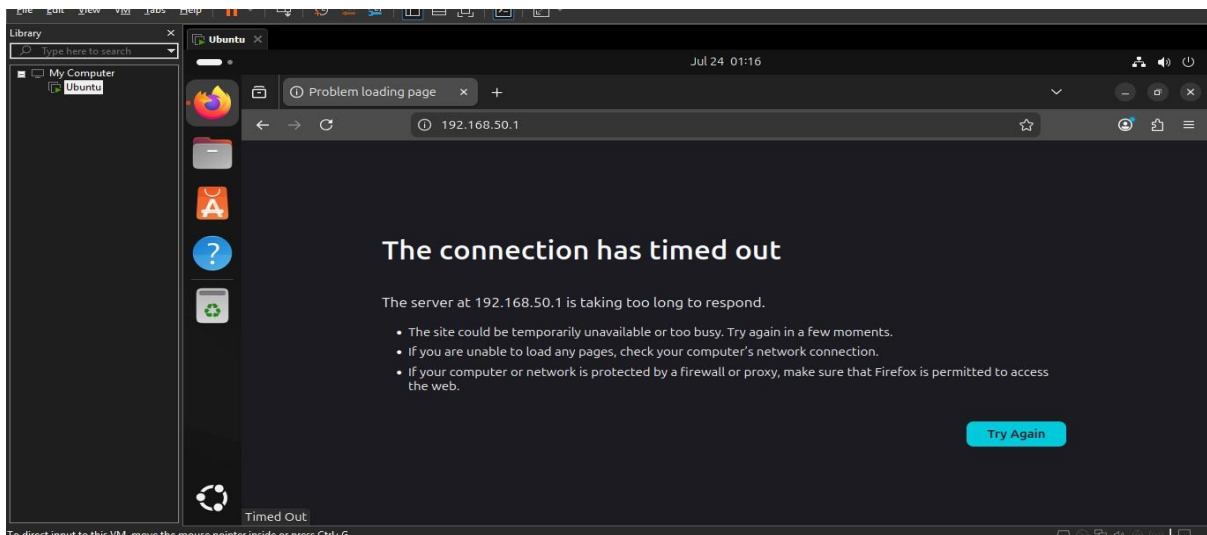
Successfully accessed the OPNsense web interface from Admin Workstation (LinuxEndpoint-FW-Test) using: <https://192.168.50.1>



Attempted to access the same web interface from a non-admin machine on the same LAN



but the browser displayed "The connection has timed out," confirming that the firewall rule successfully blocked unauthorized administrative access.



## VALIDATION SUMMARY

The implemented firewall security policies were successfully validated using the Linux

endpoint and the validation Ubuntu virtual machine. The results confirmed that the configured security controls operated as intended.

| Security Policy              | Validation Method                                       | Result                                                                                  |
|------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| GeoIP Blocking               | ping 1.0.1.1 from Linux Endpoint                        | Traffic successfully blocked                                                            |
| Web Content Filtering        | dig facebook.com and browser access                     | 0.0.0.0 returned and website inaccessible                                               |
| Administrator Access Control | Access from administrator and non-administrator systems | Administrator granted access; non-administrator received "The connection has timed out" |

## RECOMMENDATIONS:

1. **Enable Intrusion Detection:** Configure Suricata or Snort on OPNsense for additional threat detection.
2. **Regular GeoIP Database Updates:** Configure automatic updates for the **IPinfo Lite GeoIP database** to ensure accurate country-based filtering.
3. **Enable Two-Factor Authentication:** Add 2FA for administrative access to OPNsense web GUI.
4. **Configure Wazuh Alert Notifications:** Configure email or other notification mechanisms for critical security events detected by Wazuh.
5. **Implement Log Rotation:** Ensure OPNsense logs are rotated and retained for at least 30 days.
6. **Regular Rule Review:** Review firewall rules quarterly and remove outdated policies.
7. **Backup Configuration:** Schedule regular backups of OPNsense configuration.

## CONCLUSION:

The OPNsense firewall was successfully deployed, configured, and integrated with Wazuh using the Wazuh Agent. All required firewall security policies were implemented and successfully validated. GeoIP filtering effectively blocked traffic to the selected countries, DNS-based web filtering restricted access to the specified website, and administrator access control ensured that only the authorized system could access the OPNsense web management interface.

The project successfully met the objectives of Task #5 by implementing a secure firewall solution, validating firewall rule enforcement, and integrating the firewall with Wazuh for centralized monitoring. The deployment meets all Task 5 requirements and provides centralized security monitoring for the network firewall. This integration enhances the overall security posture by enabling real-time detection and alerting on malicious or unauthorized network traffic.